09/02/24 AN

Hall Ticket Number:

MR-21

Code: B414CB3S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

Introduction to Database Management Systems

(Common to CSB, CSM & CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

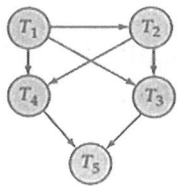
Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is a weak entity set in ER –Diagram? Quote suitable example.	2	2	1	1,2
1. b)	Explain the roles of a data base administrator.	2	2	1	1,2
1. c)	Define views and create a view from two tables.	2	2	2	1,2
1. d)	Write a query in relation algebra Account (acc_no,acc_name,bal). Find all the accounts where balance is greater than 50,000.	2	3	2	1,2
1. e)	What is trivial functional dependency?	2	2	3	1,2
1. f)	Define attribute closure with an example.	2	2	3	1,2
1. g)	What is transaction? What are the properties of transactions?	2	2	4	1,2
1. h)	Draw the transaction state diagram.	2	3	4	1,2
1. i)	What is indexing and what are the different types of indices.	2	2	5	1,2
1. j)	Differentiate between open indexing and closed indexing.	2	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	What is storage manager and query processor? Explain it with a neat diagram.	5	3	1	1,2
2. b)	Construct the Aggregation diagram and prove how it is better than ternary relationship.	5	3	1	1,2
	OR				
2. c)	Draw an E-R diagram for a university database, contains information about professors (identified by social security number, or SSN) and courses (identified by courseid). Professors teach courses; For each situation, draw an ER diagram that describes it 1. Professors can teach the same course in several semesters, and each offering must be recorded. 2. Professors can teach the same course in several semesters, 3. Every professor must teach some course. 4. Every professor teaches exactly one course (no more, no less).	5	3	2	1,2
2. d)	What are the extended ER features and explain.	5	2	1	1,2
3. a)	Consider the given schema and Write the following Relational algebra queries Customer(Customer_name, Customer_city) Borrower(Customer_name, Loan_number) Depositor(Customer_name, account_number) i) To find all customers of the bank who have an account but not a loan ii) To find all customers of the bank who have both an account and a loan iii) To find all customers of the bank who have either an account or a loan.	5	3	2	1,2
3. b)	Explain different types of joins with an example	5	2	2	1,2
	OR				

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3. c)	Write SQL Queries for the given Employee schema • List employee numbers, name sal, DA(15% OF SAL), and PF (10% of sal) • List the numbers of people and average salary in deptno 30 • List manager names and their joining dates completely spelled in alphabetical ordcr of name Employee(eno , ename, job,mgr,hirdate,sal,comm,deptno).	5	3	2	1,2
3. d)	What are the basic relational algebra operations with suitable examples.	5	2	2	1,2
4. a)	Lots(Property_id#, County_name, Lot# ,Area ,Price, Tax_rate) Apply second normal form and third normal form FD1: Property_id# → County_name Lot# Area Price Tax_rate FD2 : County_name, Lot# → Property_id#, Price, Tax_rate FD3 : County_name → Tax_rate FD4: Area → Price.	5	3	3	1,2
4. b)	Explain 4NF and 5NF with suitable examples.	5	3	3	1,2
	OR				
4. c)	Suppose that we decompose the schema R = (A, B, C, D, E) into (A, B, C) (A, D, E). Show that this decomposition is a lossless-join decomposition if the following set F of functional dependencies holds: A → BC CD → E B → D E → A	5	3	3	1,2
4. d)	Explain loss less join and loss less decomposition with examples.	5	2	3	1,2
5. a)	Explain timestamp-based protocol for the concurrency control.	5	2	4	1,2
5. b)	Explain conflict serializable schedule. Consider the precedence graph of Figure verify the corresponding schedule conflict serializable or not.  	5	2	4	1,2
	OR				
5. c)	Explain two-phase locking protocol.	5	2	4	1,2
5. d)	Consider a file system such as the one on your favourite operating system. a. What are the steps involved in creation and deletion of files, and in writing data to a file? b. Explain how the issues of atomicity and durability are relevant to the creation and deletion of files and to writing data to files.	5	3	4	1,2
6. a)	Suppose we are using extendable hashing on a file that contains records with the following search key values 2 6 9 12 16 15 19 23 14 29 Show the extendable hash structure for this file if the hash function is h(x)=x mod 8 and buckets can hold three records.	5	3	5	1,2
6. b)	Explain B+ tree insertion and deletion with an example.	5	2	5	1,2
	OR				
6. c)	Construct a B+ tree for the following set of key values. (2,3,5,7,11,17,19,23,2,31).Assume that the tree is initially empty and values are added in ascending order.	5	3	5	1,2
6. d)	Explain open hashing with suitable example.	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome